

A common viability window for self-consuming systems: from autophagic flux to dark energy — v1

Édouard Lantenois*

August 19, 2026 — v1 (v0: Aug. 18; adds the viability diagram, the strategy taxonomy, the deprivation campaign and six new systems; demotes one)

Abstract

Systems that persist by processing their own substance appear to share a dimensionless viability window in $x \equiv \tau_c/\tau_s$ (independently defined renewal requirement over self-processing time), under a protocol with four pre-registered kill conditions — all faced, none fired. Twelve counted systems spanning ~ 32 orders of magnitude of mass give $x \in [0.37, 1.95]$ (0.7 decades; central values — the liver bracket, the widest, extends to 4); controls and rate-deaths fall outside on both sides, including a natural experiment with control (fire suppression vs. the Gila Wilderness) and the clinically two-sided bone system. The *sustained* upper edges of four classes lie within 0.44 decades (two external and measured, two derived within our frameworks — labelled). A two-ingredient minimal model reproduces both bounds, the upper from the published $\sim 20\%$ maintenance calorie budget (order-level; nominal 5%). Extending the model by one growth-dilution term yields a *viability diagram* in (x, γ) whose three regions are the three observed survival strategies — residence, periodic transit (hibernation), dilution (exponential growth) — and whose dead zone ($\gamma \simeq 0, x < x_{\min}$) is observed clinically as tumor-core necrosis. A deprivation campaign to measure bare τ_c 's yields a method law (total, exogenous, lesion-free deprivations measure; mutants bound; disease blurs) and four independent literatures describing the machinery-feedback runaway the model omits. For the cosmological member, the guard-compliant variable is exactly $x = -w$: dark energy's equation of state is the viability variable, Λ is critical maintenance, the DESI crossing is the passage through criticality, and the universe's life trajectory runs from $x = 1.35$ (88% of its own sustained edge, within uncertainties) to the attractor at 0.55 — edge-proximal in youth, pack-central in age. We state plainly what would kill this.

1 Question, variable, guards

Too slow, and unprocessed damage accumulates; too fast, and the system devours itself. Is the viable band, once non-dimensionalized, common across scales — or the dimensional truism that homeostasis is $O(1)$? We define $x \equiv \tau_c/\tau_s$: τ_s the time to process one self-equivalent at the current rate; τ_c the required structural-renewal time, defined *independently* (spontaneous decay, engineering design life, dilution). A tautology guard excludes systems whose τ_c is only definable through the flux itself (epithelia; honeybee colonies; and — found by applying the guard to our own rows

*Independent researcher, Lille, France. Prepared with AI assistance (Claude, Anthropic); all load-bearing numbers verified against the cited sources; source status is labelled per row (sourced / order-of-magnitude / bounded / demoted).

— the national-accounts capital stock, *demoted* because the perpetual-inventory method derives depreciation from assumed service lives, entangling the two clocks; its repair path is engineering failure data). Two readings are declared per row, x_{self} and x_{thru} , joining physically at the alimentary ceiling where saturated intake forces excess expenditure into self-consumption [15]. Kill conditions, pre-registered: (M1) dispersion > 2 decades; (M2) controls inside as often as outside; (M3) definitional sensitivity > 1 decade; (M4, added before edge data) sustained edges scattered > 1 decade across classes.

2 Data: twelve members, two controls, exclusions

System	x	Status (abridged; full labels in the working document)
Cell, basal autophagy	0.37	mixed: basal rate order (Mortimore); increments sourced [1, 2]
Cell, starved	1.95	sourced (+3%/h) [1]
Whole mammal (thru)	0.5 (0.26–0.77)	τ_c sourced, method spread $\times 2$ declared [4]
Mitochondrial pool	0.5–1	τ_s sourced, tissue spread $\times 10$ (1.8–26 d) [5]
Fire-adapted ecosystem	0.83	sourced (FRI 5–35 y) [6]
Bone (remodeling)	0.8	both deaths sourced (see §3) [7, 8]
Liver	0.4–4	τ_s revised $\times 5$ by ^{14}C (19%/y) [9]; τ_c order-wide
City (building stock)	0.7	τ_s empirical (71 $\pm$ 28 y demolitions) [10]; τ_c bounded
Erythrocyte population	1.17	de-suspended: $\tau_c = 146$ d from storage deprivation [11]
Post-mitotic neuron	~ 0.4	order; trapped at $\gamma = 0$ (see §5)
Bacterium (exponential)	~ 0.1	below window, viable by dilution (see §5); Mandelstam check
Universe ($\beta = 2.42$)	0.85	computed; $x = -w$ (§6) [29]
<i>Controls</i> : Sun; X-ray binary	$\leq 10^{-2}$; no τ_c	consume-without-rebuild class: outside $\checkmark$
<i>Excluded by guard</i> : colony; capital	—	circular τ_c ; PIM entanglement

Table 1: $x \in [0.37, 1.95]$ over ~ 32 orders of magnitude (M1: 0.7 decades; central values — the liver bracket extends beyond). M3 executed on four systems: max displacement 0.6 decades.

3 Deaths by rate; the sustained-edge rail (M4)

Deaths bracket the window. *Low*: fire suppression ($x \rightarrow 0.15\text{--}0.3$) yields $2.9\text{--}13.6\times$ more stand-replacing fire vs. the unsuppressed Gila at $1.8\times$ — a natural experiment with control [6]; ageing neurons drift below into aggregates and a senescence–autophagy vicious circle [12]; bisphosphonate-suppressed bone accumulates unrepaired microdamage — “exponential” in the field’s own words — to atypical fractures [7]. *High*: boreal reburns ($x \sim 3\text{--}4$) fail regeneration [13]; sustained starvation-level autophagy kills dose-dependently (autosis; quantitative threshold unpublished) [14]; elevated bone resorption doubles fracture risk independently of density [8]; hierarchy offspring with $\beta > 4.35$ are GSL-unviable [30]. The sustained-edge rail — honestly labelled two external and measured (mammalian indefinite ceiling $2.5\times$ basal $\Rightarrow x \simeq 1.3$ [15, 16, 17]; boreal $\simeq 3.5$) plus two derived within our frameworks (GSL 1.53; cellular budget $\simeq 1.9$) — spans 0.44 decades: M4 passes, and the ceiling *decreases with exposure duration* ($7\times$ for days vs. $2.5\times$ indefinitely, measured [15]; transient starvation vs. sustained autosis; the GSL bounding the attractor by construction). Hibernation, the obvious would-be counterexample, is the window’s *periodic solution*: damage accrues in torpor while cold-inactivated machinery idles, survivable torpor duration is bounded, and the costly interbout arousals restore accumulated metabolites [18] — repayment/debt spanning $[0.7, 4]$ over the full

parameter grid: compatible with balance, not demonstrated closure; the homeostasis-restoration explanation and the bout-length–temperature relation belong to that literature, our contribution being only the dimensionless bookkeeping.

4 A minimal model, and the deprivation campaign

Clutter dynamics with selectivity σ give the steady state $(1 - d)(\sigma d + 1 - d) = x\sigma d$; below $x_{\min}$ (0.31–0.66 for plausible $\sigma, d_{\max}$ — illustrative) the intact fraction collapses. The budget bound, anchored to the published $\sim 20\%$ maintenance calorie fraction [3] (independently echoed by whole-body protein turnover costing $\sim 20\%$ of resting metabolism [4] — two adjacent accountings, noted as scale consistency only), gives $x_{\max} = 1.85$, with the observed autotrophic edge 5% *beyond* bankruptcy — expected physics, order-level agreement ($\pm 30\%$ real). A campaign to measure bare τ_c ’s by *deprivation* yielded: two clean (stored blood: total, exogenous, ex vivo $\Rightarrow \tau_c = 146$ d, de-suspending the erythrocyte row; bisphosphonates: pharmacological suppression, quantified); one epidemiological (bone’s high side); one partial (PINK1-KO accumulates dysfunction in 4–9 months [19] but backup mitophagy persists [20]: upper bound only, $x \leq 4$ –6, tension declared); one runaway (abandoned buildings measure the breach–water feedback’s escape time, 3–15 y [21], not a bare constant); one impure (chronic liver disease confounds injury with arrest — yet establishes the healthy senescence clock exceeds a century: “a cirrhotic liver shows more senescence than a healthy centenarian’s” [22]). *Method law*: a good τ_c -meter requires total, exogenous, lesion-free deprivation — storage and drugs measure, mutants bound, disease blurs. And the machinery feedback our minimal model deliberately omits is now described independently by four literatures: neuronal senescence–autophagy, “exponential” bone microdamage, the storage lesion’s ATP–glycolysis spiral, and the breach–water runaway of abandoned buildings.

5 The viability diagram: three strategies, one dead zone

Adding one growth-dilution term ($\gamma = g\tau_c$, new intact material) generalizes the window to a plane: $d' = (1 - d) - x\sigma d/(\sigma d + 1 - d) - \gamma d$, with lower boundary $x_{\min}(\gamma)$ falling to zero at $\gamma_{\text{crit}} = (1 - d_{\max})/d_{\max} \simeq 0.43$ (analytic; Fig. 1). The three observed survival strategies are the plane’s regions: *residence* ($\gamma \approx 0$, x in window: neuron — trapped there, its literature’s stated vulnerability [12] — cell, bone, universe); *dilution* ($\gamma > \gamma_{\text{crit}}$, x free to collapse: exponential bacteria at $\gamma \sim 14$, proliferative tumor rims; consistency check: stationary-phase proteolysis rises severalfold, Mandelstam’s classic result); *periodic transit* (hibernation: a vertical limit cycle). The dead zone ($\gamma = 0$, $x < x_{\min}$) is observed clinically: tumor-core necrosis. The universe is declared a structurally distinct case: its “growth” (expansion) dilutes the *maintained* substance rather than the damage — effectively $\gamma < 0$, off the biological plane. Parameters $\sigma, d_{\max}$ remain post-hoc plausible: the diagram is an extension of the same illustrative model, its value structural (one equation, three strategies, one predicted dead region), not numerical.

6 The identity $x = -w$, and the universe’s life in the window

With the guard-compliant degradation for the cosmological member being *dilution* ($\tau_c = 1/3H$, $\tau_s = t/\beta$), the viability variable is $x = \beta/(3Ht) = -w$ exactly — general under the declared convention of a pressureless bare substance. Λ is critical maintenance ($x = 1$); phantom is the autotrophic side; quintessence the clutter side; the DESI crossing is the passage through criticality ($z \simeq 0.46$), thermodynamically a smooth crossover [28]. Formal kin: $w = -1 + \dot{S}_h/(3HS_h)$ on

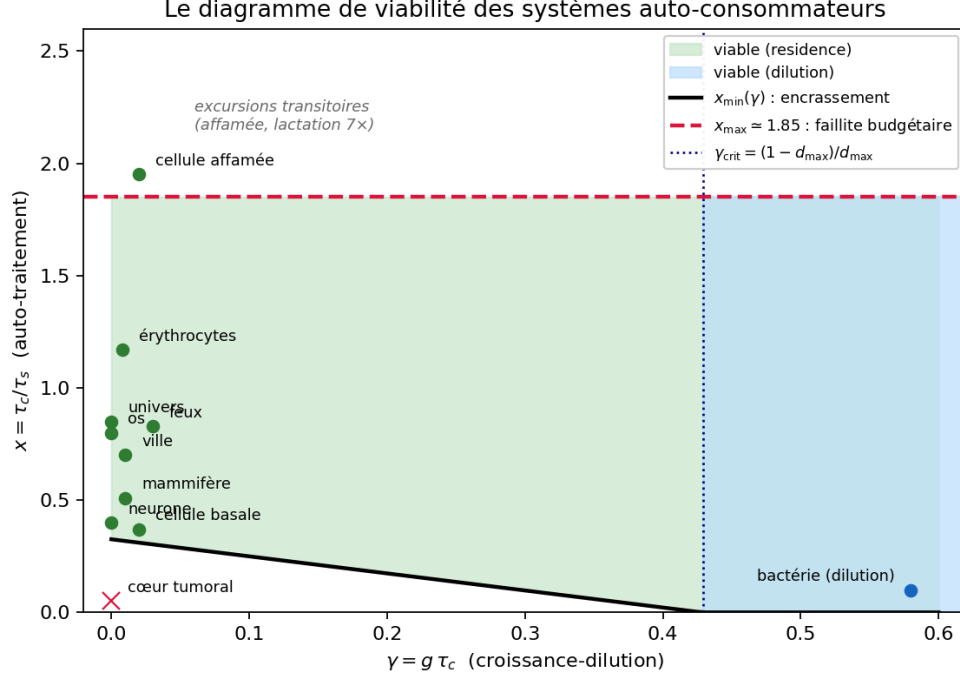


Figure 1: The viability diagram (x, γ) : clutter boundary $x_{\min}(\gamma)$, budget line $x_{\max} \simeq 1.85$, dilution region beyond $\gamma_{\text{crit}} = 0.43$, systems placed (deaths as red crosses). Display note: the bacterium is drawn at the plot's right edge; its true $\gamma \approx 14$ lies far off-scale. Illustrative parameters $\sigma = 5$, $d_{\max} = 0.7$.

the horizon side [26]; classical fluid-entropy positivity ($w \geq -1$) [27] vs. our more permissive total-entropy GSL bound ($w \geq -1.53$) — a contentful difference. Window bounds become w -bounds: viability $\Leftrightarrow w \in [-1.53, -0.35]$ (Ht fixed at best fit, $\sim 10\text{--}15\%$ map effect), observed dark energy at -0.85 , mid-window; and the universe's trajectory runs from $x = 1.35$ deep in the past ($\sim 88\%$ of its own sustained edge, below it within uncertainties) through criticality to today's 0.85 and the attractor's 0.55 : an entire cosmic history inside the sustained window, edge-proximal in youth, pack-central in age. Both faces declared: the identity is partly definitional; its content is the derived bounds and the shared vocabulary.

7 Ancestry, open items, kill list

Ancestors: rate of living [23, 24] (x as its maintenance-normalized correction); disposable soma [25] ($x_{\max}$ as its quantitative form); prudent parent [17]. Open: the unpublished quantitative autosis threshold; a total-deprivation mitochondrial τ_c ; more external edge classes for M4 (the rail holds only two external points); external review. Kills: a guard-compliant class far outside $[0.3, 2]$; new sustained edges scattering the rail beyond a decade; failure of the duration tier where measurable; or a demonstration that the surviving rows share a hidden definitional convention, as capital did. Each is a concrete assignment.

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